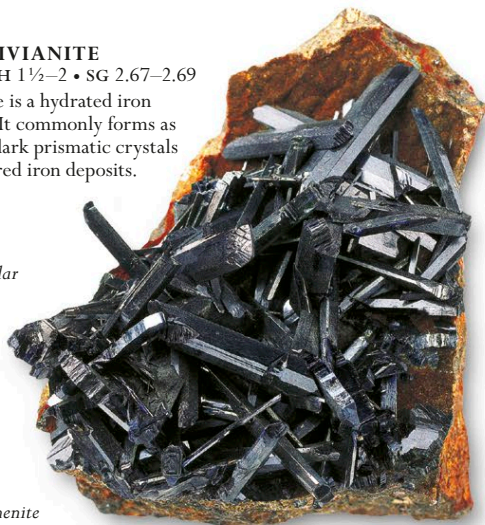


VIVIANITE

Monoclinic • H 1½–2 • SG 2.67–2.69

Vivianite is a hydrated iron phosphate. It commonly forms as clusters of dark prismatic crystals in altered iron deposits.

radiating acicular
(needle-shaped)
crystals



libethenite
crystal

**LIBETHENITE**Orthorhombic
H 4 • SG 3.97

Libethenite is a light to dark green copper phosphate hydroxide. It forms in the upper oxidized zone of copper deposits.

MONAZITE

Monoclinic • H 5–5½ • SG 5–5.5

Phosphate minerals containing either cerium, lanthanum, or neodymium are all referred to as monazite. They are mined for the various elements.

**BRAZILIANITE**

Monoclinic • H 5½ • SG 2.98

First discovered in Brazil, this sodium aluminium phosphate hydroxide is yellow or greenish and forms in cavities in granitic pegmatites.

**LAZULITE**Monoclinic
H 5½–6 • SG 3.12–3.24

A relatively rare, semi-precious blue gemstone, this magnesium aluminium phosphate hydroxide occurs in metamorphic and igneous rocks.



bipyramidal
crystal

VANADINITE

Hexagonal • H 2½–3 • SG 6.88

This relatively rare lead vanadate chloride forms crystals in altered lead deposits. It is an important source of vanadium, used in steel alloys.

**VANADATES**

Vanadates are formed by the combination of metallic elements and the vanadate radical (VO_4). This group contains many rare minerals, which may be dense and brightly coloured. Vanadates often form when hydrothermal veins are altered by permeating fluids. Most vanadates have no commercial value; however, carnotite is an important source of uranium.

powdery crust on sandstone

CARNOTITE

Monoclinic • H 2 • SG 4.75

Generally occurring as powdery yellow crusts in uranium deposits, radioactive carnotite is a hydrated vanadate of potassium and uranium.

**TYUYAMUNITE**Orthorhombic
H 1½–2 • SG 3.57–4.35

This rare hydrated vanadate of calcium and uranium looks similar to carnotite, and also occurs in altered uranium deposits.

**ARSENATES**

Arsenates are mostly rare minerals composed of metallic elements and the arsenate radical (AsO_3 or AsO_4). They generally have a low hardness. Many arsenates are brightly coloured – adamite is yellow or green, and clinoclase is green or blue. This group of minerals form in a variety of geological situations, but many arsenates occur in altered metal deposits.

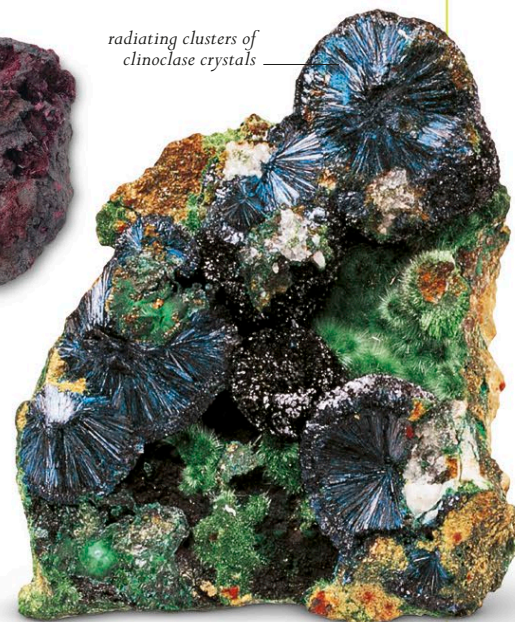
ERYTHRITE

Monoclinic • H 1½–2½ • SG 3.06

This hydrated cobalt arsenate forms purple-pink crystals or coatings. Excellent examples occur in Canada and Morocco.



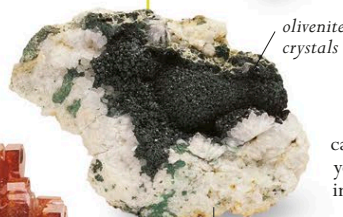
radiating clusters of
clinoclase crystals

**BAYLDONITE**Monoclinic
H 4½ • SG 5.24–5.65

This hydrous arsenate of copper and lead is usually found as green or yellow crusts in altered hydrothermal veins.

**OLIVENITE**Monoclinic
H 3 • SG 4.46

Olivenite is a copper arsenate hydroxide. It can be greenish, brownish, yellow, or grey and occurs in altered copper deposits.



olivinite
crystals

quartz

**MIMETITE**

Hexagonal • H 3½–4 • SG 7.24

Unusual, barrel-shaped crystals characterize this lead arsenate chloride, although other forms also exist. Mimetite occurs in altered lead deposits.

CHALCOPHYLLITE

Trigonal • H 2 • SG 2.67–2.69

Bright blue-green chalcophyllite is a hydrated copper aluminium arsenate sulfate. It forms in oxidized copper deposits.

